Chapter 11
Reactor Kinetics and Point Kinetics

The point kinetics model can be obtained directly from the space- and time-dependent transport equations. However, these equations are too complicated to be of any practical application. The diffusion approximation, obtained by keeping only the PI terms of the spherical harmonic expansion in the angular variable of the directional flux, is frequently used in neutronic analysis. This chapter discusses reactor characteristics that change because of changing reactivity. A basic approach using a minimum of mathematics has been followed. Emphasis has been placed on distinguishing between prompt and delayed neutrons and showing relationships among reactor variables, k_{eff}, period, neutron density, and power level.

11.1 Time-Dependent Diffusion Equation

For the analysis of point reactor kinetics, the situation arises where one needs to consider the transient changes resulting from the departure of the reactor condition from the critical state. This results under the following conditions as:

1. Startup.
2. Shutdown.
3. Accidental disturbances in the course of what is intended to be steady-state operation.

The power produced during a reactor transient is one of the most significant factors, which is determining the degree of damage that can happen from an accident within reactor.

The time-dependent power production is related to the effective multiplication factor k_{eff} and the prompt and delayed neutron properties through the reactor kinetics equations. The spatial distribution of the reactor neutron flux can be ignored in lieu of an emphasis on its time behavior. The reactor is viewed as a point; hence, the

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terminology of point reactor kinetics arises. In this regard, a distinction must be made between the behaviors of the prompt and delayed neutrons.

Point kinetics is very interesting because of the apparent simplicity of the resulting differential equations. The method is very frequently used, but the underlying difficulties in obtaining the parameters are hidden. In spite of all this, many inherent characteristics of the dynamics of nuclear cores can be deduced from these equations. In addition, these same equations provide a tool for the analysis, the comparison, and the practical implementation of various numerical schemes that may eventually be used in more complex situations. An integration technique that does not pass the test of point kinetics will certainly not be used in space-time kinetics. Point kinetics can thus play the role of an experimental bench before expensive problems are attempted with more advanced methods [1].

The driving factor, behind point kinetics, is to separate the flux into two factors: the first one being a shape function depending both on space and time and a second factor depending only on time in the following fashion:

$$\phi\left(\vec{\boldsymbol{r}},t\right)=S\left(\vec{\boldsymbol{r}},t\right)T(t) \tag{11.1}$$

Note that this equation for the flux does not involve any approximation and that the equality is maintained. However, the shape function $S\left(\vec{\boldsymbol{r}},t\right)$ depends both on space and on time in this approach. We now introduce a column vector of weight functions as:

$$W\left(\vec{\boldsymbol{r}}\right)=\begin{bmatrix} W_1\left(\vec{\boldsymbol{r}}\right) \\ W_1\left(\vec{\boldsymbol{r}}\right) \\ \vdots \\ W_G\left(\vec{\boldsymbol{r}}\right) \end{bmatrix} \tag{11.2}$$

whose function will be to give *rise* to general equations. In effect, Eq. 11.1 presents a degree of arbitrariness in the choice of $S\left(\vec{\boldsymbol{r}},t\right)$ and of $T(t)$; only the product of the two variables needs to be specified. We will use $W\left(\vec{\boldsymbol{r}}\right)$ to introduce normalization constraints, which it will obey at all times during a transient condition. Specifically, we define:

$$T(t)=\left\langle [W]^T[\mathrm{v}]^{-1}[\phi]\right\rangle, \tag{11.3}$$

and it *follows* that Eq. 11.1 and the shape function $S\left(\vec{\boldsymbol{r}},t\right)$ must obey the following constraint, by substituting Eq. 11.1 into Eq. 11.3:

$$\left\langle [W]^T[\mathrm{v}]^{-1}\left[S\left(\vec{\boldsymbol{r}},t\right)\right]\right\rangle=1 \tag{11.4}$$

where *the* symbol $\langle\rangle$ means spatial integration over the whole domain of the nuclear core.

Note that the factor $T(t)$ in Eq. 11.3 is called the amplitude function and the shape function $S(\vec{\boldsymbol{r}},t)$ represents in some sense the total number of neutrons in the reactor, but this number depends on the weight function. As the constraint on $S(\vec{\boldsymbol{r}},t)$ does not depend on time, the shape function may change in time, but its integral is time independent. Thus, $T(t)$ itself represents the neutron population change in the reactor.

We can now seek for a differential equation for the time-dependent variable $T(t)$ by replacing $\phi(\vec{\boldsymbol{r}},t)$ by the product of $S(\vec{\boldsymbol{r}},t)T(t)$ in the *space-time kinetics* equations, by *pre-multiplying the resulting equations* by $W(\vec{\boldsymbol{r}})]^T$, and by integrating over the whole core volume. Thus using Eq. 11.4, we obtain the following results:

$$\left\langle \left[W(\vec{\boldsymbol{r}})\right]^T \left\{ \begin{array}{c} \left\langle [W]^T[\mathrm{v}]^{-1}[S]\right\rangle \dfrac{\partial T}{\partial t} \\ \nabla\cdot[D]\,\vec{\nabla}\,[S]-[\Sigma][S] \\ +\left((1-\beta)[\chi^p]+\displaystyle\sum_{i=1}^{D}\beta_i\left[\chi_i^d\right]\right)\left[\mathrm{v}\Sigma_f\right]^T[S] \end{array} \right\} \right\rangle T(t)$$

$$-\left\langle \left[W(\vec{\boldsymbol{r}})\right]^T \sum_{i=1}^{D}\beta_i\left[\chi_i^d\right]\left[\mathrm{v}\Sigma_f\right]^T[S] \right\rangle T(t)+\sum_{i=1}^{D}\lambda_i\left\langle [W]^T\left[\chi_i^d\right]C_i\right\rangle \tag{11.5}$$

where in order to *conform* to certain conventions, we have added and subtracted the term in:

$$\sum_{i=1}^{D}\beta_i\left[\chi_i^d\right]\left[\mathrm{v}\Sigma_f\right]^T[S] \tag{11.6}$$

Very similar operations are performed on the precursor equations that we pre-multiply by $[W]^T\left[\chi_i^d\right]$ before integrating over space to get the following relationship:

$$\frac{\partial}{\partial t}\left\langle [W]^T\left[\chi_i^d\right]C_i\right\rangle=\left\langle [W]^T\beta_i\left[\chi_i^d\right][S]\right\rangle T-\lambda_i\left\langle [W]^T\left[\chi_i^d\right]C_i\right\rangle \tag{11.7}$$

We now *define* the following quantities as:

$$C_i(t)=\frac{\left\langle [W]^T\left[\chi_i^d\right]C_i\right\rangle}{\left\langle [W]^T[\mathrm{v}]^{-1}[S]\right\rangle} \tag{11.8}$$

Now, we first define another quantity that is known as the *mean neutron generation time* as $\Lambda(t)$ so we can write:

$$\Lambda(t)=\frac{\left\langle [W]^T[\mathrm{v}]^{-1}[S]\right\rangle}{\left\langle [W]^T\left\{(1-\beta)[\chi^P]+\sum_{i=1}^{D}\beta_i[\chi_i^d]\right\}\left[\mathrm{v}\Sigma_f\right]^T[S]\right\rangle} \tag{11.9}$$

where *again* $\Lambda(t)$ is defined as:

$\Lambda(t)\equiv\frac{1}{k}\equiv$ *mean generation time* between birth of neuron and subject absorption inducing fission.

Note that if $k{\sim}1$, then $\Lambda(t)$ is essentially just the prompt neutron lifetime ℓ.

Now, if we introduce the quantity of $\beta_i(t)$, as the following form:

$$\beta_i(t)=\beta_i\frac{\left\langle [W]^T\left[\chi_i^d\right]\left[\mathrm{v}\Sigma_f\right]^T[S]\right\rangle}{\left\langle [W]^T\left\{(1-\beta)[\chi^P]+\sum_{i=1}^{D}\beta_i[\chi_i^d]\right\}\left[\mathrm{v}\Sigma_f\right]^T[S]\right\rangle} \tag{11.10}$$

then we can *write*:

$$\beta(t)\approx\sum_{i=1}^{D}\beta_i(t) \tag{11.11}$$

Now if we define a very important quantity known as the *reactivity* and represented by $\rho(t)$, which essentially measures the deviation of core multiplication from its critical value, where $k=1$, then we can write the following equation by utilizing what we have shown in Eq. 11.5:

$$\rho(t)=\frac{\left[W(\vec{\boldsymbol{r}})\right]^T\left\{\begin{array}{l}\nabla\cdot[D]\stackrel{\rightarrow}{\nabla}[S]-[\Sigma][S]\\+\left((1-\beta)[\chi^P]+\sum_{i=1}^{D}\beta_i\left[\chi_i^d\right]\right)\left[\mathrm{v}\Sigma_f\right]^T[S]\end{array}\right\}}{\left\langle [W]^T\left\{(1-\beta)[\chi^P]+\sum_{i=1}^{D}\beta_i[\chi_i^d]\right\}\left[\mathrm{v}\Sigma_f\right]^T[S]\right\rangle} \tag{11.12}$$

Another form of reactivity factor $\rho(t)$ is presented by Duderstadt and Hamilton [2] in the following:

$$\rho(t)=\frac{k(t)-1}{k(t)}=\mathit{reactivity} \tag{11.13}$$

Note that Eq. 11.13 explicitly indicates criticality factor k as a function of time t; hence reactivity factor ρ may be a function of time as well.

Now that we have definitions in Eq. 11.10 and Eq. 11.12, the *space-time kinetics equation* becomes:

$$\begin{cases} \dfrac{dT}{dt} = \dfrac{[\rho(t) - \beta]}{\Lambda(t)} T + \sum_{i=1}^{D} \lambda_i C_i \\ \dfrac{dC_i}{dt} = \dfrac{\beta_i(t)}{\Lambda(t)} - \lambda_i C_i \quad i = 1, \cdots, 6 \end{cases} \tag{11.14}$$

Note that Eq. 11.14 presents a set of seven coupled ordinary differential equations, in time, which describe both the time dependence of the neutron population in the reactor and the decay of the delayed neutron precursor [2].

11.2 Derivation of Exact Point Kinetics Equations (EPKE)

The most accurate description of the neutron behavior in a near critical nuclear reactor is given by the three-dimensional, energy-, angle-, and time-dependent form of the Boltzmann transport equation given by:

$$\begin{aligned} &\frac{1}{\mathrm{v}(E)} \frac{d\phi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)}{dt} + \vec{\boldsymbol{\Omega}} \cdot \nabla \phi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) + \Sigma_t(\vec{\boldsymbol{r}}, E, t)\phi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) = q(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) \\ &+ \int_{E'} \int_{\vec{\boldsymbol{\Omega}}'} \Sigma_s(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}' \to E, \vec{\boldsymbol{\Omega}}, t)\phi(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t) dE' d\vec{\boldsymbol{\Omega}}' \\ &+ \chi(E, t) \int_{E'} \int_{\vec{\boldsymbol{\Omega}}'} \frac{\upsilon(E')}{4\pi} \Sigma_f(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t)\phi(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t) dE' d\vec{\boldsymbol{\Omega}}' \end{aligned} \tag{11.15}$$

where fission has been assumed to be isotropic in the laboratory reference frame. In addition, the following quantities are defined:

$\phi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)$ is the time-dependent angular flux density (the number of neutrons at position $\vec{\boldsymbol{r}}$, moving in direction $\vec{\boldsymbol{\Omega}}$ with energy E, at time t)
$\mathrm{v}(E)$ is the velocity of neutrons with energy E
$\sum_t(\boldsymbol{r}, E, t)$ is the macroscopic material total cross section for neutrons of energy E, at position $\vec{\boldsymbol{r}}$, and time t
$q(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)$ is the source of neutrons at position $\vec{\boldsymbol{r}}$, energy E, and moving in direction $\vec{\boldsymbol{\Omega}}$ at time t
$\Sigma_s(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}' \to E, \vec{\boldsymbol{\Omega}}, t)$ is the macroscopic material cross section for scattering from direction $\vec{\boldsymbol{\Omega}}'$ and energy E' to direction $\vec{\boldsymbol{\Omega}}$ and energy E at position $\vec{\boldsymbol{r}}$ and time t
$\chi(E, t)$ is the time-dependent yield of fission neutrons at energy E

$\upsilon(E')$ is the total yield of neutrons because of fission caused by a neutron of energy E'

$\Sigma_f\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)$ is the macroscopic fission cross section for neutrons of energy E' at position $\vec{\boldsymbol{r}}$ and time t

Fission has been assumed isotropic in the laboratory reference frame.

Since not all neutrons are produced at the moment of fission, the source term must be broken up into an independent source term and a delayed neutron source term. This can be accomplished as follows. Let:

$$\chi(E,t)\int_{E'}\int_{\vec{\boldsymbol{\Omega}}'}\frac{v}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)\phi\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)dE'd\vec{\boldsymbol{\Omega}}'$$
$$= \chi_p(E)(1-\beta)\int_{E'}\int_{\vec{\boldsymbol{\Omega}}'}\frac{v}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)\phi\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)dE'd\vec{\boldsymbol{\Omega}}' + \sum_{i=1}^{M}\chi_{id}(E)\lambda_i C_i\left(\vec{\boldsymbol{r}}, t\right) \tag{11.16}$$

where a separate *energy* spectrum has been prescribed for prompt and delayed neutrons and the precursor concentrations have been introduced as $C_i\left(\vec{\boldsymbol{r}}, t\right)$.

This equation can theoretically be solved by applying straightforward finite difference methods in all variables (seven for the most general case). However, this is a very large computational task and has never really been accomplished for the most general case. Usually the number of dimensions has been reduced, by taking advantage of symmetry in the spatial dimensions and the lack of a strong angular dependence in the energy-dependent flux. Even so, the energy-dependent diffusion equations in more than one spatial dimension present a formidable computation task. It has been found that this level of effort is not required for a significant number of reactor kinetics problems and a heuristic point kinetics equation approach is very accurate. The objective of the following derivation is to make explicit the approximations inherent in the point kinetics equations and therefore in some way provide an indication of their adequacy in untried circumstances.

With this goal in mind, it is useful to factor the time-dependent angular flux density into two parts:

$$\phi\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right) = \varphi\left(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t\right)\cdot P(t) \tag{11.17}$$

where $P(t)$ will *hopefully* contain the major time dependence and ϕ will not be a strong function of time. In order to unambiguously perform this splitting, or factorization, a rule for sorting out $P(t)$ is required. The most useful rule is:

$$\int_{\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}} W\left(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}\right)\frac{1}{\mathrm{v}(E)}\varphi\left(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t\right) = 1.0 \tag{11.18}$$

where $W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})$ is some *arbitrary* weighting function yet to be selected. Multiplying the transport equation by $W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})$ and integrating over all space, energies, and angles, the following equation can be obtained:

$$\frac{d}{dt}\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\frac{1}{\mathrm{v}(E)}\varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)P(t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19a}$$

$$+\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\;\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)P(t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19b}$$

$$+\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\Sigma_t(\vec{\boldsymbol{r}}, E, t)\varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)P(t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19c}$$

$$=\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})q(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19d}$$

$$+\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\int_{E',\vec{\boldsymbol{\Omega}}}\Sigma_s(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}' \to E, \vec{\boldsymbol{\Omega}}, t)\varphi(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t)P(t)dE'd\vec{\boldsymbol{\Omega}}'d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19e}$$

$$+\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\chi_p(E)(1-\beta)\int_{E',\vec{\boldsymbol{\Omega}}}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t)\varphi(\vec{\boldsymbol{r}}, E', \vec{\boldsymbol{\Omega}}', t)$$
$$P(t)dE'd\vec{\boldsymbol{\Omega}}'d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19f}$$

$$+\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\sum_{i=1}^{M}\chi_{id}(E)\lambda_i C_i(\vec{\boldsymbol{r}}, t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \tag{11.19g}$$

The individual terms in this equation are identified because it will be convenient to reduce them one at a time.

Consider first Eq. 11.19a. Removing $P(t)$ from the phase space integrals gives:

$$\text{Eq.11.19a} = \begin{cases} \dfrac{d}{dt}P(t)\displaystyle\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\frac{1}{\mathrm{v}(E)}\varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} + \\ P(t)\dfrac{d}{dt}\displaystyle\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}})\frac{1}{\mathrm{v}(E)}\varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t)d\,\vec{\boldsymbol{r}}\;dEd\,\vec{\boldsymbol{\Omega}} \end{cases}$$

Applying the normalization condition reduces this term to:

$$\text{Eq.11.19a} = \frac{d}{dt}P(t)\cdot 1.0 + P(t)\cdot\frac{d}{dt}1.0 = \frac{d}{dt}P(t)$$

This is the major motivation for choosing the normalization condition in the form it was specified.

The next step is to define the equilibrium fission source term. The equilibrium fission source is composed of the prompt and delayed components. At equilibrium, or the critical condition, the spatial dependence of the delayed spectrum is the same as the prompt spectrum spatial dependence. The difference between the equilibrium spectrum and the conventional spectrum is that the delayed part of the neutron emission has a different energy spectrum. For the equilibrium or critical condition fission source, we have:

$$\begin{aligned}
&\chi(E)\int_{E,\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)\phi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)dE'd\vec{\boldsymbol{\Omega}}'\\
&=\chi_p(E)(1-\beta)\int_{E',\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)\phi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)dE'd\vec{\boldsymbol{\Omega}}'\\
&+\sum_{i=1}^{M}\chi_{id}(E)\beta_i\int_{E',\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)\phi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\right)dE'd\vec{\boldsymbol{\Omega}}'
\end{aligned}\tag{11.20}$$

Therefore, to incorporate the equilibrium fission source, we must add and subtract the delayed fission source components from the transport equation. When this is done and the terms are added to *T6*, it becomes:

$$\text{Eq.11.19f} = \left\{\begin{aligned}
&\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}}W\left(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}\right)\{\chi_p(E)(1-\beta)\int_{E',\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)\\
&\varphi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)P(t)dE'd\vec{\boldsymbol{\Omega}}'\\
&+\sum_{i=1}^{M}\chi_{id}(E)\beta_i\int_{E',\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)\varphi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)P(t)\\
&-\sum_{i=1}^{M}\chi_{id}(E)\beta_i\int_{E',\vec{\boldsymbol{\Omega}}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)\varphi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)P(t)\}\\
&d\,\vec{\boldsymbol{r}}\,dEd\,\vec{\boldsymbol{\Omega}}
\end{aligned}\right.\tag{11.21}$$

Now define an equilibrium reduced fission source, *FS,* as:

$$\int_{\vec{r},E,\vec{\Omega}} W\left(\vec{r},E,\vec{\Omega}\right)\left\{\chi_p(E)(1-\beta)+\sum_{i=1}^{M}\chi_{id}(E)\beta_i\right\}\int_{E',\vec{\Omega}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{r},E',\vec{\Omega}',t\right)$$
$$\varphi\left(\vec{r},E',\vec{\Omega}',t\right)dE'd\vec{\Omega}'d\,\vec{r}\;dEd\,\vec{\Omega} \tag{11.22}$$

And dividing every term by *FS* will give the following equation:

$$\{\text{Eq.}11.19\text{a}+\text{Eq.}11.19\text{b}+\text{Eq.}11.19\text{c}\}/FS=\{\text{Eq.}11.19\text{d}+\text{Eq.}11.19\text{e}$$
$$+\text{Eq.}11.19\text{f}+\text{Eq.}11.19\text{g}\}/FS$$

Identifying $1/FS$ as Λ and rearranging the terms give:

$$\Lambda Eq.11.19\text{a}=\Lambda\{\text{Eq.}11.19\text{e}+\text{Eq.}11.19\text{f}-\text{Eq.}11.19\text{b}\}-\text{Eq.}11.19\text{c}$$
$$+\Lambda Eq.11.19\text{d}+\Lambda Eq.11.19\text{g}$$

Now consider Eq. 11.19d and define a reduced independent source as:

$$Q(t)=\int_{\vec{r},E,\vec{\Omega}'} W\left(\vec{r},E,\vec{\Omega}\right)q\left(\vec{r},E,\vec{\Omega},t\right)d\,\vec{r}\;,dE,d\,\vec{\Omega} \tag{11.23}$$

Define $Z_i(t)$ as:

$$Z_i(t)=\int_{\bar{r},E,\overline{\Omega}} W\left(\vec{r},E,\vec{\Omega}\right)C_i\left(\vec{r},t\right)\chi_{id}(E)d\,\vec{r}\;dEd\,\vec{\Omega} \tag{11.24}$$

The transport equation can now be written as:

$$\Lambda\frac{dP(t)}{dt}=\Lambda\{\text{Eq.}11.19\text{f}+\text{Eq.}11.19\text{e}+-\text{Eq.}11.19\text{b}-\text{Eq.}11.19\text{c}\} \tag{11.25}$$

Consider Eq. 11.19f again and write it as:

$$\text{Eq. }11.19\text{f}=FS^{*}P(t)+\text{Eq. }11.19\text{f} \tag{11.26}$$

where:

$$\text{Eq.}11.19\text{f}=-\sum_{i=1}^{M}\beta_i\chi_{id}(E)\int_{E',\vec{\Omega}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{r},E'\vec{\Omega}',t\right)\varphi\left(\vec{r},E',\vec{\Omega}',t\right)P(t)dE'd\vec{\Omega}' \tag{11.27}$$

Then, defining:

$$\beta = -\frac{\text{Eq. } 11.19\text{f}'\Lambda}{P(t)} \tag{11.28}$$

and:

$$\rho = \frac{\{\text{Eq.}11.19\text{e} + FS^* P(t) - \text{Eq.}11.19\text{b} - \text{Eq.}11.19\text{c}\}\Lambda}{P(t)} \tag{11.29}$$

gives:

$$\Lambda \frac{dP(t)}{dt} = (\rho - \beta)P(t) + \Lambda Q(t) + \Lambda \sum_{i=1}^{M} \lambda_i Z_i(t) \tag{11.30}$$

or:

$$\frac{dP(t)}{dt} = \frac{\rho - \beta}{\Lambda} P(t) + Q(t) + \sum_{i=1}^{M} \lambda_i Z_i(t) \tag{11.31}$$

This is essentially the power equation for the point kinetics description of reactor behavior.

Now consider the precursor equations given by:

$$\frac{dC_i(\vec{\boldsymbol{r}}, t)}{dt} = \beta_i \int_{E,\vec{\boldsymbol{\Omega}}} \frac{\mathrm{v}(E)}{4\pi} \Sigma_f(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) \varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) P(t) dE d\vec{\boldsymbol{\Omega}} - \lambda_i C_i(\vec{\boldsymbol{r}}, t) \tag{11.32}$$

Multiplying these equations by:

$$\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}) \chi_{id}(E) d\vec{\boldsymbol{r}}\, dE d\vec{\boldsymbol{\Omega}} \tag{11.33}$$

will give:

$$\begin{aligned} \frac{dZ_i(\vec{\boldsymbol{r}}, t)}{dt} = {} & \beta_i \int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} \chi_{id}(E) W(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}) \int_{E',\vec{\boldsymbol{\Omega}}'} \frac{\mathrm{v}(E)}{4\pi} \Sigma_f(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) \varphi(\vec{\boldsymbol{r}}, E, \vec{\boldsymbol{\Omega}}, t) \\ & P(t) dE' d\vec{\boldsymbol{\Omega}}' d\vec{\boldsymbol{r}}\, dE d\vec{\boldsymbol{\Omega}} - \lambda_i C_i(\vec{\boldsymbol{r}}, t) \end{aligned} \tag{11.34}$$

Remembering that:

$$\beta = \frac{-\text{Eq.11.19f}'\Lambda}{P(t)}$$
$$= \frac{\sum_{i=1}^{M}\beta_i \int_{\bar{r},E,\overline{\Omega}} \chi_{id} W \int_{E',\overline{\Omega}'} \frac{\mathrm{v}(E')}{4\pi}\Sigma_f \varphi P(t) dE' d\overline{\Omega}' d\vec{\boldsymbol{r}}\, dEd\overline{\Omega}}{P(t)\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W\left\{\chi_p(1-\beta) + \sum_{i=1}^{M}\chi_{id}\beta_i\right\}\int_{E',\overline{\Omega}'} \frac{\mathrm{v}(E')}{4\pi}\Sigma_f \varphi dE' d\vec{\boldsymbol{\Omega}}' d\vec{\boldsymbol{r}}\, dEd\vec{\boldsymbol{\Omega}}} \tag{11.35}$$

The precursor equations become:

$$\frac{dZ_i(t)}{dt} = \frac{\beta_i}{\Lambda}P(t) - \lambda_i Z_i(t) \tag{11.36}$$

The "exact" point kinetics equations are then:

$$\frac{dP(t)}{dt} = \frac{\rho-\beta}{\Lambda}P(t) + \sum_{i=1}^{M}\lambda_i Z_i(t) \tag{11.37}$$

$$\frac{dP(t)}{dt} = \frac{\rho-\beta}{\Lambda}P(t) + \sum_{i=1}^{M}\lambda_i Z_i(t) \tag{11.38}$$

These equations are exact if all of the parameters in them are computed according to the formulas given above. However, to apply the formulas given above exactly would be as difficult a task as integrating the original transport equation directly. The real advantage of deriving the "exact" equations in this manner is to rigorously prescribe how to compute the quantities used to predict kinetic behavior. When approximations are introduced in computing these quantities, at least there is a standard to compare against in order to understand the impact of the approximations.

Consider the definition of ρ. The complete expression for ρ is given by:

$$\rho = \frac{\{\text{Eq.11.19e} + FS^{*}P(t) - \text{Eq.11.19b} - \text{Eq.11.19c}\}\Lambda}{P(t)} \tag{11.39}$$

or:

$$\left\{\begin{array}{l}\int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W \int_{E',\vec{\boldsymbol{\Omega}}'} \Sigma_s\left(E',\vec{\boldsymbol{\Omega}}' \to E,\vec{\boldsymbol{\Omega}}'\right)\varphi dE' d\vec{\boldsymbol{\Omega}}' d\vec{\boldsymbol{r}}\, dEd\vec{\boldsymbol{\Omega}} + \\ \int_{\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}}} W\left\{\chi_p(1-\beta) + \sum_{i=1}^{M}\chi_{id}\beta_i\right\}\int_{E',\vec{\boldsymbol{\Omega}}'} \frac{v}{4\pi}\Sigma_f \phi dE' d\overline{\Omega}' d\vec{\boldsymbol{r}}\, dEd\overline{\Omega}\end{array}\right. \tag{11.40a}$$

or we find ρ to be as follows:

$$\rho = \frac{-\int_{\vec{r},E,\vec{\Omega}} W\,\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi d\,\vec{r}\,dEd\,\vec{\boldsymbol{\Omega}} - \int_{\overline{r},E,\overline{\Omega}} W\Sigma_t\varphi d\,\vec{r}\,dEd\,\vec{\boldsymbol{\Omega}}}{\int_{\vec{r},E,\vec{\Omega}} W\left\{\chi_p(1-\beta)+\sum_{i=1}^{M}\chi_{id}\beta_i\right\}\int_{E',\vec{\Omega}}\frac{v}{4\pi}\Sigma_f\varphi dE'd\vec{\boldsymbol{\Omega}}'d\,\vec{r}\,dEd\,\vec{\boldsymbol{\Omega}}} \tag{11.40b}$$

where the functional dependence of the major variables has been suppressed. The notation can be further simplified by defining:

$$F\varphi = \int_{E,\vec{\Omega}} \frac{\mathrm{v}(E')}{4\pi}\Sigma_f\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)\varphi\left(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t\right)dE'd\vec{\boldsymbol{\Omega}}' \tag{11.41}$$

$$\chi = \chi_p(1-\beta)+\sum_{i=1}^{M}\chi_{id}\beta_i \tag{11.42}$$

and suppressing most of the variable dependence. Then:

$$\rho = \frac{-\int_{\vec{r},E,\vec{\Omega}} W\left(\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi+\Sigma_t\varphi-\chi F\varphi-\Sigma_s\varphi\right)}{\int_{\vec{r},E,\vec{\Omega}} W\chi F\varphi} \qquad \beta = \frac{\sum_{i=1}^{M}\beta_i\int W\chi_{id}F\varphi}{\int_{\vec{r},E,\vec{\Omega}} W\chi F\varphi} \tag{11.43}$$

$$\Lambda = \frac{1}{\int_{\vec{r},E,\vec{\Omega}} W\chi F\varphi} \qquad Q(t) = \int_{\vec{r},E,\vec{\Omega}} Wq \qquad Z_i(t) = \int_{\vec{r},E,\vec{\Omega}} W\chi_{id}C_i \tag{11.44}$$

Note that for a critical reactor, the static eigenvalue equation is given by:

$$\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi+\Sigma_t\varphi = \chi F\varphi+\Sigma_s\varphi \tag{11.45}$$

It is also worth noting at this time that ϕ is not the diffusion flux, but is actually a reduced transport flux and definitely can contain anisotropic components that may be necessary to evaluate in order to correctly calculate the reactivity. However, the analysis that we have gone through will apply to both transport models and diffusion theory models.

11.3 The Point Kinetics Equations

Now that we have derived the point kinetics equations (PKE), we should investigate what they say about nuclear reactor behavior. We begin by noting that form a homogeneous set of ordinary differential equations. As we know the solution to a homogeneous set of equations only exists for certain parameters called eigenvalues. We can write the equations as:

$$
\begin{aligned}
\frac{dP}{dt} &= \frac{\rho-\beta}{\Lambda}P + \lambda_1 z_1 + \lambda_2 z_2 + \lambda_3 z_3 + \lambda_4 z_4 + \lambda_5 z_5 + \lambda_6 z_6 \\
\frac{dz_1}{dt} &= \frac{\beta_1}{\Lambda}P \quad - \lambda_1 z_1 \\
\frac{dz_2}{dt} &= \frac{\beta_2}{\Lambda}P \qquad\quad - \lambda_2 z_2 \\
\frac{dz_3}{dt} &= \frac{\beta_3}{\Lambda}P \qquad\qquad\quad - \lambda_3 z_3 \\
\frac{dz_4}{dt} &= \frac{\beta_4}{\Lambda}P \qquad\qquad\qquad\quad - \lambda_4 z_4 \\
\frac{dz_5}{dt} &= \frac{\beta_5}{\Lambda}P \qquad\qquad\qquad\qquad\quad - \lambda_5 z_5 \\
\frac{dz_6}{dt} &= \frac{\beta_6}{\Lambda}P \qquad\qquad\qquad\qquad\qquad\quad - \lambda_6 z_6
\end{aligned}
\tag{11.46}
$$

And if we assume a form for P and each z_i that incorporates an eigenvalue ω such that:

$$
P(t) = P_0 e^{\omega t} \quad \frac{dP(t)}{dt} = \omega P(t) \quad Z_i(t) = Z_{i0} e^{\omega t} \quad \frac{dZ_i(t)}{dt} = \omega Z_i(t) \tag{11.47}
$$

then each of the $Z_i(t)$ equations can be solved for Z_i to get:

$$
(\omega + \lambda_i) z_i(t) = \frac{\beta_i}{\Lambda} P(t) \quad z_i(t) = \frac{\beta_i}{\Lambda(\omega + \lambda_i)} P(t) \tag{11.48}
$$

Then:

$$
\omega P(t) = \frac{\rho-\beta}{\Lambda} P(t) + P(t) \sum_{i=1}^{6} \frac{\lambda_i \beta_i}{\Lambda(\omega + \lambda_i)} \tag{11.49}
$$

And dividing out $P(t)$ gives:

$$\begin{cases} \sum_{i=1}^{6} \frac{\lambda_i \beta_i}{\omega + \lambda_i} = \rho - \sum_{i=1}^{6} \beta_i \frac{\omega + \lambda_i}{\omega + \lambda_i} + \sum_{i=1}^{6} \beta_i \frac{\lambda_i}{\omega + \lambda_i} = \rho - \omega \sum_{i=1}^{6} \frac{\beta_i}{\omega + \lambda_i} \\ \rho = \omega \left(\Lambda + \sum_{i=1}^{6} \frac{\beta_i}{\omega + \lambda_i} \right) \end{cases} \tag{11.50}$$

There are seven roots for w in this inhour equation.

Inhour Equation

The relationship between a step change in reactivity (ρ) and the resulting reactor period (T), allowing for the delayed neutrons, can be derived in the form of an equation such as:

$\rho = \frac{\ell}{kT} + \sum_{i=1}^{6} \frac{\beta_i}{1 + \lambda_i T}$ where:

β_i is delayed neutron fraction for group i
λ_i is decay constant of delayed neutron group $i(s^{-1}) = \ln 2/t_{1/2}$
ℓ is prompt neutron lifetime (s)
T is reactor period.

This equation is often called the reactivity equation or the inhour equation, although strictly speaking this is not the inhour equation proper. (The term inhour comes from expressing reactivity in the units of inverse hours. The inhour unit is then that reactivity will make the stable reactor period equal to 1 h.) These are numerous different formulations of the reactivity equation, but this one will suffice for the purposes of this book.

Now consider the One Delay Family model. The inhour equation in this case becomes:

$$\left[\begin{array}{l} \rho = \omega \left(\Lambda + \frac{\beta}{\omega + \lambda} \right) \\ (\omega + \lambda)\rho = \omega\Lambda(\omega + \lambda) + \omega\beta \\ \omega \frac{\rho - \beta}{\Lambda} + \frac{\lambda\rho}{\Lambda} = \omega^2 + \lambda\omega \\ \omega^2 + \left(\lambda - \frac{\rho - \beta}{\Lambda} \right) \omega - \frac{\lambda\rho}{\Lambda} = 0 \end{array} \right. \tag{11.51}$$

Note that for most cases, if $\rho < \beta$, $\lambda \ll (\rho - \beta)/\Lambda$, so λ can be neglected. Then:

$$\omega = \frac{\rho-\beta}{2\Lambda} \pm \sqrt{\left(\frac{\rho-\beta}{2\Lambda}\right)^2 + \frac{\lambda\rho}{\Lambda}} = \frac{\rho-\beta}{2\Lambda} \pm \frac{\rho-\beta}{2\Lambda}\sqrt{1+\frac{4\lambda\rho}{\Lambda\left(\frac{\rho-\beta}{\Lambda}\right)^2}} \tag{11.52}$$

And noting that $\sqrt{1+x} = 1 + \frac{x}{2} + \ldots.$ for small x:

$$\begin{cases} \omega_1 = \dfrac{\rho-\beta}{\Lambda} + \dfrac{\rho-\beta}{\Lambda}\dfrac{\lambda\rho}{\Lambda\left(\frac{\rho-\beta}{\Lambda}\right)^2} = \dfrac{\rho-\beta}{\Lambda} + \dfrac{\lambda\rho}{\rho-\beta} \approx \dfrac{\rho-\beta}{\Lambda} \\ \omega_2 = -\dfrac{\lambda\rho}{\rho-\beta} = \dfrac{\lambda\rho}{\beta-\rho} \end{cases} \tag{11.53}$$

Therefore, the general solution for a step reactivity change is:

$$P(t) = P_{01}e^{\frac{\rho-\beta}{\Lambda}t} + P_{02}e^{\frac{\lambda\rho}{\beta-\rho}t} \tag{11.54}$$

We will find the unknown coefficients P_{01} and P_{02} from the initial conditions. Continuity of the power and the derivative of the power give:

$$P_0 = P_{01} + P_{02} \tag{11.55a}$$

$$\begin{aligned} \frac{dP}{dt}(0) &= \frac{\rho-\beta}{\Lambda}P_0 + \lambda z_0 \\ &= \frac{\rho-\beta}{\Lambda}P_0 + \lambda\frac{\beta}{\lambda\Lambda}P_0 \\ &= P_0\left(\frac{\rho}{\Lambda}\right) \\ &= \omega_1 P_{01} + \omega_2 P_{02} \end{aligned} \tag{11.55b}$$

$$\begin{aligned} P_0\left(\frac{\rho}{\Lambda}\right) &= \frac{\rho-\beta}{\Lambda}(P_0 - P_{02}) + \frac{\lambda\rho}{\beta-\rho}P_{02} \\ P_0\frac{\beta}{\Lambda} &= \left(\frac{\lambda\rho}{\beta-\rho} - \frac{\rho-\beta}{\Lambda}\right)P_{02} \\ P_{02} &\approx \frac{\beta}{\beta-\rho}P_0 \\ P_{01} &= \frac{-\rho}{\beta-\rho}P_0 \end{aligned} \tag{11.55c}$$

This gives the complete solution for a step insertion of reactivity as:

$$P(t) = P_0\left\{\frac{\beta}{\beta-\rho}e^{\frac{\lambda\rho}{\beta-\rho}t} - \frac{\rho}{\beta-\rho}e^{\frac{\rho-\beta}{\Lambda}t}\right\} \tag{11.56}$$

Now that we are here, next we have dynamic and static reactivity, which is the subject of the next section here.

11.4 Dynamic Versus Static Reactivity

An often discussed concept is the difference between dynamic and static reactivity. There are a large number of ways of computing reactivity which is being used in the field today. The simplest, most straightforward, and least accurate is the eigenvalue difference method. That is, two reactor configurations are mocked up as critical systems. An eigenvalue, K, is found that makes each of them critical, and the difference between these two eigenvalues is assumed to be the same as the reactivity defined above. In the limit of small reactivity, this will be approximately correct; however, as the time-dependent flux shape changes in a reactor, it becomes more and more difficult to evaluate reactivity based on a hypothetical eigenvalue.

A more accurate method would be to use the formalism defined above for the time-dependent case but use a time-independent shape function. This static function usually corresponds to the shape function for an exactly critical reactor, $k = 1.0$. In this case, the weighting function, W, can be chosen to make this approximate reactivity as accurate as possible by letting it equal the ADJOINT function for the exactly critical reactor. If the critical ADJOINT function is used, then the reactivity will be insensitive to first-order changes in the shape function; thus, the static reactivity will be as accurate an approximation to the actual reactivity as is possible.

The most accurate method of computing the reactivity is to use the actual time-dependent shape function in the integral. Of course, this is much more difficult to do, as somehow the time dependence has to be calculated. It is also important to note that for this case, the choice of W is somewhat arbitrary. A constant value of 1.0 could be chosen, but the ADJOINT function for an exactly critical reactor is normally used in hopes of mitigating some of the errors introduced in calculating the time-dependent shape function.

The three definitions are:

1. *Eigenvalue difference reactivity*

$$\rho = \frac{k_1 - k_0}{k_1 k_0} \tag{11.57}$$

2. *Static reactivity*

$$\rho = \frac{\int_{\vec{r},E,\vec{\Omega}} \varphi^* \left(\vec{\Omega} \cdot \nabla\varphi + \Sigma_t\varphi - \chi F\varphi - \Sigma_s\varphi\right) d\,\vec{r}\,dEd\,\vec{\Omega}}{\int_{\vec{r},E,\vec{\Omega}} \phi^* \chi F\varphi d\,\vec{r}\,dEd\,\vec{\Omega}} \tag{11.58}$$

where the *ADJOINT* function for the critical configuration is used as the weighting function and the fluxes are obtained from a static criticality calculation for the current configuration of the reactor assembly.

3. *Dynamic reactivity*

$$\rho = \frac{\int_{\vec{r},E,\vec{\Omega}} \phi^* \left(\overline{\Omega} \cdot \nabla\varphi + \Sigma_t\varphi - \chi F\varphi - \Sigma_s\varphi\right) d\,\vec{r}\,dEd\,\vec{\Omega}}{\int_{\vec{r},E,\vec{\Omega}} \varphi^* \chi F\varphi d\,\vec{r}\,dEd\,\vec{\Omega}} \tag{11.59}$$

where the ADJOINT function for the critical configuration is used as the weighting function and the fluxes are obtained from a time-dependent calculation for the shape function. The difference between the most accurate approach for static reactivity, ADJOINT weighting, and the true dynamic reactivity is the use of the time-dependent shape function in the dynamic reactivity case. So now, consider how to calculate this shape function.

11.5 Calculating the Time-Dependent Shape Function

Given that the dynamic reactivity is the actual quantity required in the point kinetics equations, an integrodifferential equation can be derived for its calculation. This equation can then be approximated in many ways in order to obtain solutions that are useful for particular problems. Substituting the factored flux into the transport equation and dividing by $P(t)$ give:

$$\begin{aligned}
&\frac{1}{\mathrm{v}(E)}\left\{\frac{d\varphi(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}},t)}{dt}+\varphi(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}},t)\frac{dP(t)}{dt}/P(t)\right\}\\
&+\overline{\Omega}\cdot\nabla\varphi(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}},t)+\Sigma_t(\vec{\boldsymbol{r}},E,t)\varphi(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}},t)=\\
&\int_{E',\overline{\Omega}'}\Sigma_s(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}'\to\boldsymbol{E},\vec{\boldsymbol{\Omega}},t)\varphi(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t)dE'd\vec{\boldsymbol{\Omega}}'\\
&+\chi_p(1-\beta)\int_{E',\overline{\Omega}'}\frac{\mathrm{v}(E')}{4\pi}\Sigma_f(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t)\varphi(\vec{\boldsymbol{r}},E',\vec{\boldsymbol{\Omega}}',t)dE'd\vec{\boldsymbol{\Omega}}'\\
&+\frac{1}{P(t)}\left\{q(\vec{\boldsymbol{r}},E,\vec{\boldsymbol{\Omega}},t)+\sum_{i=1}^{M}\chi_{id}(E)\beta_i\lambda_iZ_i(\vec{\boldsymbol{r}},t)\right\}
\end{aligned}\tag{11.60}$$

Of course, this equation is more complicated than the original transport equation. It would be much harder to integrate it directly for any particular problem. However, by performing the factorization and writing the equation in this manner, simplifications are possible depending on the problem.

In shorthand notation this equation can be written as:

$$\frac{1}{\mathrm{v}(E)}\frac{d\varphi}{dt}+\frac{1}{\mathrm{v}(E)}\varphi\frac{dP}{dt}/P+\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi+\Sigma_t\varphi=\Sigma_s\varphi+\chi_pF\varphi+\frac{1}{P}\left\{q+\sum_{i=1}^{M}\chi_{id}\beta_i\lambda_iZ_i\right\}\tag{11.61}$$

Rearranging the terms gives:

$$\frac{1}{\mathrm{v}(E)}\frac{d\varphi}{dt}+\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi+\Sigma_t\varphi=\Sigma_s\varphi+\chi_pF\varphi+S/P\tag{11.62}$$

where:

$$S=q+\sum_{i=1}^{M}\chi_{id}\beta_i\lambda_iZ_i-\frac{1}{v}\varphi\frac{dP}{dt}$$

This looks almost identical to the complete transport equation except for the S/P term. If P is large and S small (fast transients), this term can be neglected, and the shape function equation is essentially the same as the original transport equation. Thus, it would appear that no simplification has been achieved, and for the most general case, this is indeed true. However, in many cases, the time dependence of the shape function is considerably smaller than the time dependence of the angular flux density, and significantly larger time steps can be taken in its integration.

11.6 Point Kinetics Approximations

If the time dependence of the shape function is completely neglected, all quantities can be evaluated for the critical reactor. This is often considered the least accurate of the methods for dealing with the shape function. However, it will be accurate enough for many applications.

11.6.1 Level of Approximation to the Point Kinetics Equations

As we have seen, the point kinetics equations (PKE) are approximations to the full spatial and time-dependent behavior of a nuclear reactor. For very many cases of reactor behavior, they are a quite accurate approximation and are a more than adequate approximation to allow complete control of the reactor and characterization of its behavior.

However the PKE are still too complicated to easily explain some behavior and derive physical data about the reactor from observed measurements. Therefore, we are going to look at several levels of approximation the PKE mainly based on how accurately or detailed we treat the delayed neutron phenomena. The following levels of approximation have applications as noted and are frequently used to describe reactor transients.

- No delayed neutron approximation (*NDN*)
- Prompt kinetics (*PK*)
- Constant delayed source (*CDS*)
- Precursor accumulation (*PA*)
- One delay family (*ODF*)
- Prompt jump approximation [zero lifetime] (*PJA* or *ZL*)

11.7 Adiabatic Approximation

If the reactor configuration at any point in time is considered to be static, the problem becomes a homogenous one. This is exactly the problem most design codes are set up to calculate. (However, the eigenvalue K must now be introduced in order to obtain a solution. Note this eigenvalue will also compensate for the fact that the reduced equation is based on $\chi_p(E)$ rather than $\chi(E)$). Note that the eigenfunction obtained may not in this case correspond to any real reactor flux shape. And even if the critical ADJOINT solution is used, the accuracy of this technique can be questioned. In addition, since this method requires the computation of an eigenvalue-eigenfunction solution every time it is desired to update the shape function, it can be computationally expensive.

11.8 Adiabatic Approximation with Pre-computed Shape Functions

One method that has been proposed for overcoming this computation obstacle is the use of pre-computed shape functions and their mixing as time goes on to approximate the actual shape function. This is one of those tricks, which will usually work if you know the answer to your problem before you try to solve it, but for "extrapolation" cases, "let the analyst beware." In general, it is not recommended, and accuracy limits cannot be specified.

11.9 Quasi-Static Approximation

In this approximation the shape function is integrated on a much longer time interval than the power function, $P(t)$. The "inhomogeneous" source term S/P can be included or not included. The form used will depend on which terms are relevant for a particular problem and part of the transient is being evaluated. Normally a less than second-order implicit scheme will be used for this integration.

That means either an Euler's backward method or a trapezoidal rule integration method. The Euler's backward method is much faster to compute and therefore has generally been the method of choice.

For the no source (S/P) case, this would be written as:

$$\left\{\frac{1}{\mathrm{v}(E)\Delta t}+\vec{\boldsymbol{\Omega}}\cdot\nabla+\Sigma_t-\Sigma_s-\chi_p F\right\}\varphi_2=\frac{1}{\mathrm{v}(E)\Delta t}\varphi_1 \tag{11.63}$$

and the flux shape at the start of the time interval becomes the source for the shape at the end of the time interval. If parts of the S/P source are to be included, they are time differenced and added to the equation on the right- and left-hand sides as appropriate.

11.10 Zero-Dimensional Reactors

Now consider a diffusion theory model of a zero-dimensional reactor. We will base this on an infinite slab, but the analysis is the same for any zero-dimensional model. The flux and adjoint solutions are:

$$\begin{aligned}\phi(E,z) &= \sum_{g=1}^{NOG} \varphi^{g} \cos Bz \\ \phi^{*}(E,z) &= \sum_{g=1}^{NOG} \varphi^{*g} \cos Bz\end{aligned} \tag{11.64}$$

Now consider evaluating some of the terms in the point kinetics equations. Start with the generation time:

$$\Lambda = \frac{1}{\int_{-H/2}^{H/2} \left[\sum_{g=1}^{NOG} \varphi^{g} \cos Bz \left\{ \chi_{p}^{g}(1-\beta) + \sum_{i=1}^{M} \beta_{i} \chi_{id}^{g} \right\} \sum_{g=1}^{NOG} v \sum\nolimits_{f}^{g} \varphi^{g} \cos Bz \right] dz} \tag{11.65}$$

There are a couple of items to note about this definition. First of all this is the only term where the absolute normalization of φ^{*g} is important. In all other terms, there will be a φ^{*g} on the bottom and on the top of the parameter definition, so only the relative values will be important. In order to get the absolute normalization, we must go back to the definition of the factorization:

$$\begin{aligned}&\int_{\vec{r},E\vec{\Omega}} W\left(\vec{r}, E\, \vec{\Omega}\right) \frac{1}{\mathrm{v}(E)} \varphi\left(\vec{r}, E\, \vec{\Omega}, t\right) = 1.0 = \sum_{g=1}^{NOG} \frac{\varphi^{*g}\varphi^{g}}{\mathrm{v}^{g}} \int_{-H/2}^{H/2} \cos^{2} Bz dz \\ &= \frac{\pi}{2} \sum_{g=1}^{NOG} \frac{\varphi^{*g}\varphi^{g}}{\mathrm{v}^{g}} = 1.0\end{aligned} \tag{11.66}$$

This will give us the absolute normalization of the weighting function or adjoint function. The weighting function can be written as some constant W_0 times the normalized adjoint function. The inner product of the normalized adjoint and flux divided by the group velocity can then be computed, and the normalizing constant becomes 1 over the inner product:

$$\begin{aligned}W(E,z) &= W_{0} \sum_{g=1}^{NOG} \varphi^{*g} \cos Bz \\ W_{0} &= \frac{1}{\frac{\pi}{2} \sum_{g=1}^{NOG} \frac{\varphi^{*g}\varphi^{g}}{\mathrm{v}^{g}}}\end{aligned} \tag{11.67}$$

Then we can define a χ^g as:

$$\chi^g = \chi_p^g(1-\beta) + \sum_{i=1}^{M} \beta_i \chi_{id}^g \tag{11.68}$$

Also note that the second summation in the denominator is completed before it is multiplied by the first two terms, so we can sum the fission source once and for all and use it in evaluating all of the other integrals.

Thus, we can write a simplified generation time computation as:

$$\begin{aligned}\Lambda &= \frac{1}{W_0 \frac{\pi}{2}\left(\sum_{g=1}^{NOG} \varphi^{*g}\chi^g\right)(FS)} \\ &= \frac{1}{W_0 \frac{\pi}{2}\left(\sum_{g=1}^{NOG} \varphi^{*g}\chi^g\right)\left(\sum_{g=1}^{NOG} \mathrm{v}\Sigma_f^g \varphi^g\right)}\end{aligned} \tag{11.69}$$

Now we can calculate $\beta_{\text{effective}}$ as:

$$\beta = \frac{\sum_{i=1}^{M} \beta_t \int_{-H/2}^{H/2} \sum_{g=1}^{NOG} W_0 \varphi^{*g} \cos Bz \chi_{id}^g \sum_{g=1}^{NOG} \mathrm{v}\Sigma_f^g \varphi^g \cos Bz dz}{W_0 \frac{\pi}{2}\left(\sum_{g=1}^{NOG} \varphi^{*g}\chi^g\right)\left(\sum_{g=1}^{NOG} \mathrm{v}\Sigma_f^g \varphi^g\right)} \tag{11.70}$$

Note that in this case, the W_0 divides out on the top and bottom as does the $\pi/2$ factor and the total fission source. The formula becomes:

$$\beta = \frac{\sum_{i=1}^{M} \beta_i \sum_{g=1}^{NOG} \varphi^{*g}\chi_{id}^g}{\sum_{g=1}^{NOG} \varphi^{*g}\chi^g} \tag{11.71}$$

Finally, it is worth making some comments about the reactivity term. We have for the diffusion model:

$$\begin{cases} \rho = -\dfrac{\displaystyle\int_{\vec{r},E\vec{\boldsymbol{\Omega}}} W\left(\vec{\boldsymbol{\Omega}}\cdot\nabla\varphi + \Sigma_t\varphi - \chi F\varphi - \Sigma_s\varphi\right)}{\displaystyle\int_{\vec{r},E\vec{\boldsymbol{\Omega}}} W\chi F\varphi} \\ \rho = -\dfrac{\displaystyle\int_{-H/2}^{H/2}\sum_{g'=1}^{NOG} W_0\varphi^{*g}\left(D^g B^2\varphi^g + \Sigma_\gamma^g\varphi^g - \chi^g\sum_{g'=1}^{NOG}\mathrm{v}\Sigma_f^{g'}\varphi^{g'} - \sum_{g'=g-1}^{1}\Sigma_s^{g'\to g}\varphi^{g'}\right)\cos^2 Bz dz}{W_0\dfrac{\pi}{2}\left(\displaystyle\sum_{g=1}^{NOG}\varphi^{*g}\chi^g\right)\left(\displaystyle\sum_{g=1}^{NOG}\mathrm{v}\Sigma_f^g\varphi^g\right)} \\ \rho = -\dfrac{\displaystyle\sum_{g'=1}^{NOG}\varphi^{*g}\left(D^g B^2\varphi^g + \Sigma_\gamma^g\varphi^g - \chi^g\sum_{g'=1}^{NOG}\mathrm{v}\Sigma_f^{g'}\phi^{g'} - \sum_{g'=g-1}^{1}\Sigma_s^{g'\to g}\varphi^{g'}\right)}{\left(\displaystyle\sum_{g=1}^{NOG}\varphi^{*g}\chi^g\right)\left(\displaystyle\sum_{g=1}^{NOG}\mathrm{v}\Sigma_f^g\varphi^g\right)} \end{cases} \tag{11.72}$$

Note that the numerator contains the exact material properties and flux shape functions. If we make the linear perturbation approximation, this can be rewritten as:

$$\rho = -\frac{\displaystyle\sum_{g'=1}^{NOG}\varphi_0^{*g}\left(\delta D^g B^2\varphi_0^g + \delta\Sigma_\gamma^g\varphi_0^g - \chi^g\sum_{g'=1}^{NOG}\delta\mathrm{v}\Sigma_f^{g'}\varphi_0^{g'} - \sum_{g'=g-1}^{1}\delta\Sigma_s^{g'\to g}\varphi_0^{g'}\right)}{\left(\displaystyle\sum_{g=1}^{NOG}\varphi^{*g}\chi^g\right)\left(\displaystyle\sum_{g=1}^{NOG}\mathrm{v}\Sigma_f^g\varphi^g\right)} \tag{11.73}$$

where basically we have subtracted off the critical equation and said that the product of two small changes is negligible compared to a single small change.

Problems

Problem 11.1 Thermal Reactor Example

Consider a thermal reactor example. The following results were obtained for a six-group model of the Advanced Reactivity Measurement Facility reactor: a small 93% enriched aluminum, slab core reactor. The model is based on a bare core version of this reactor shaped into a cube with a 51.09 cm side dimension. This gave a buckling of $B^2 = 0.011344/\text{cm}^2$. Note that this is a fairly small reactor.

Consider the following results:

Group	Upper energy	Velocity	$v\Sigma_f$	Flux	Adjoint	χ_p	χ_d	χ
1	17.33 MeV	4.24 (+9) cm/s	0.000643	0.03691	0.122862	0.05258	1.27(−6)	0.05221
2	4.97. MeV	2.24 (+9) cm/s	0.000492	1.00383	0.531294	0.52040	0.050525	0.51711
3	1.35 MeV	8.66 (+8) cm/s	0.000433	2.03885	0.695888	0.41086	0.803196	0.41361
4	111.0 keV	1.93 (+8) cm/s	0.000854	0.59341	0.849079	0.01608	0.145344	0.01699
5	3.35 keV	5.95 (+6) cm/s	0.01863	1.89784	0.875391	8.0(−5)	0.000933	8.6(−5)
6	0.100 eV	2.20 (+5) cm/s	0.18219	1.00000	1.000000	0.0	0.0	0.0

Calculate the normalization factor.
Calculate $\beta_{\text{effective}}$ and β_{physica} for this problem.

Problem 11.2 The thermal absorption cross section for Xe-135 at 2200 meters is 2.7E+6 barns. Based on the bounds developed in class, what is the maximum absorption cross section that could be expected at 0.0253 eV for this nuclide? Estimate the total cross section at 2.0 MeV also.

Problem 11.3 Determine the thermal lifetime or diffusion time of neutron in the bare critical reactor where $L^2 = 57.3$ and $B^2 = 0.0051\ \text{cm}^{-2}$ and consisting of beryllium and uranium-235 in the atomic ratio of 10^4 to 1.

Problem 11.4 The reactivity in a steady-state reactor, in which the neutron generation time is 10^{-3} s, is suddenly made 0.0022 positive; assuming one group of delayed neutrons, determine the subsequent change of neutron flux with time. Compare the stable period with that which would result had no delayed neutrons. Use the following equation for variation of the neutron density with time:

$$n \approx n_0 \left[\frac{\beta}{\beta - \rho} e^{\frac{\lambda \rho t}{e^{\beta - \rho}}} - \frac{\rho}{\beta - \rho} e^{-\frac{(\beta - \rho)t}{l^*}} \right] \tag{11.74a}$$

In addition, for the stable reactor period, assuming one (average) group of delayed neutron is given by:

$$T_p \approx \frac{\beta - \rho}{\lambda \rho} \tag{11.75b}$$

Assume on average, for one group of delayed neutron, λ is 0.08 s^{-1}; ρ is given as 0.0022 and neutron generation time l^* as 10^{-3} s; β for uranium-235 is 0.0065.

Problem 11.5 The prompt neutron lifetime in a reactor moderated by heavy water is 5.7×10^{-4} s. For a reactivity of 0.00065, express the reactor period (a) in second, (b) in hours (Ih), using the relationship between reactor period T_p, and in hours (Ih) as $T_p \approx 3600/\text{Ih}$. What is the reactivity in dollar units? Assume for very long periods and very small reactivity the following equation applies and $(\beta/\lambda) = 0.084$:

$$T_p \approx \frac{\beta}{\lambda\rho} \tag{11.76a}$$

where:

T_p is reactor period
β is total fraction of delayed neutrons = 0.0065
λ is radioactive decay constant
ρ is reactivity

In addition, the unit that is called "the dollar," is defined by:

$$\text{Reactivity in dollars} \equiv \frac{\rho}{\beta}$$

Problem 11.6 As part of one-group point kinetics equation, if we assume a case for when only one group of delayed neutron is assumed, the results of solution provide a polynomial in the denominator as part of general solution, when we assume source term is constant and equal to S_0 with two roots of s_1 and s_2 of the following mathematical notation:

$$s_{1,2} = \frac{-\left(\frac{\beta}{\Lambda} - \frac{\rho_0}{\Lambda} + \lambda\right) \pm \sqrt{\left(\frac{\beta}{\Lambda} - \frac{\rho_0}{\Lambda} + \lambda\right)^2 + 4\frac{\lambda\rho_0}{\Lambda}}}{2} \tag{11.77a}$$

where:

β is fractional yield of the delayed neutrons
Λ is average neutron generation time
ρ_0 is reactivity for constant source S_0
λ is decay constant.

Then the neutron flux can be obtained as:

$$\begin{aligned} n(t) &= n_e x(t) + n_e \\ &= n_e\left\{1 + \left(\frac{\rho_0}{\Lambda} + \frac{S_0}{n_e}\right)\left[\frac{\lambda}{s_1 s_2} + \frac{s_1 + \lambda}{s_1(s_1 + s_2)}e^{s_1 t} + \frac{s_2 + \lambda}{s_2(s_2 - s_1)}e^{s_2 t}\right]\right\} \end{aligned} \tag{11.78b}$$

where:

$n(t)$ is neutron density as function of time t
n_e is neutron density at equilibrium

Assuming one group of delayed neutrons, determine the subsequent change of neutron flux with time. Additionally, the reactivity is taking place in a steady-state thermal reactor with no external neutron source, in which the neutron generation time $\Lambda = 10^{-3}$ s is suddenly made 0.0022 positive and the following data applies: $\lambda = 0.08\ \mathrm{s}^{-1}$ and $\beta = 6.5 \times 10^{-3}$.

Problem 11.7 A critical reactor operated during a long period of time with the mean weighted number of neutrons equal to $n_e = 10^6$. Suddenly a source of neutrons with mean weighted yield equal to $S_0 = 10^6\ \mathrm{s}^{-1}$ was introduced into the reactor. Find the number of neutrons as a function of time, $n(t)$. In calculations assume one group of delayed neutrons with $\lambda = 0.1\ \mathrm{s}^{-1}$ and $\beta = 6.4 \times 10^{-3}$ and the mean generation time $\Lambda = 10^{-3}$ s.

Problem 11.8 An under-critical reactor operated during a long period of time with the reactivity equal to $\rho_0 = -9\beta$. A source of neutrons with a mean weighted yield equal to $S_0 = 10^6\ \mathrm{s}^{-1}$ was present in the reactor. Suddenly the source of neutrons was removed from the reactor. Find the number of neutrons as a function of time, n(t). In calculations assume one group of delayed neutrons with $\lambda = 0.1\ \mathrm{s}^{-1}$ and $\beta = 6.5 \times 10^{-3}$ and the mean generation time $\Lambda = 10^{-3}$ s. Note that the equilibrium number of neutrons before the removal of the source is given by $n_e = -S_0\Lambda/\rho_0$.

References

1. J. Koclas, *Neutronic Analysis of Reactor*, Nuclear Engineering Department at Ecole Polytechnique de Montreal, February 1998
2. J. Duderstadt, L. Hamilton, *Nuclear Reactor Analysis* (John Wiley Publishing Company, 1976)